

Surath Majumdar | Senior Platform Engineer

Driving platform reliability, DevEx & automated governance across hybrid and cloud-native environments

✉ surath.majumdar@gmail.com | ☎ (470) 209-1438 | 🏠 Newbury Park, CA

🔗 <https://www.linkedin.com/in/surath-majumdar/> | 🐙 <https://github.com/surath-majumdar/>

Project Brief:

Architecting an Internal Developer Platform (IDP) to Scale DevEx

Architect & Lead Engineer: Surath Majumdar

Domain: Platform Engineering, Developer Experience (DevEx), Infrastructure Automation, DevSecOps

Core Technologies: Backstage (DevHub), Terraform, AWS (Secrets Manager, IAM), Kubernetes (Kind, Cilium), GitOps (ArgoCD), Open Policy Agent (OPA), AsyncAPI

1. Executive Summary

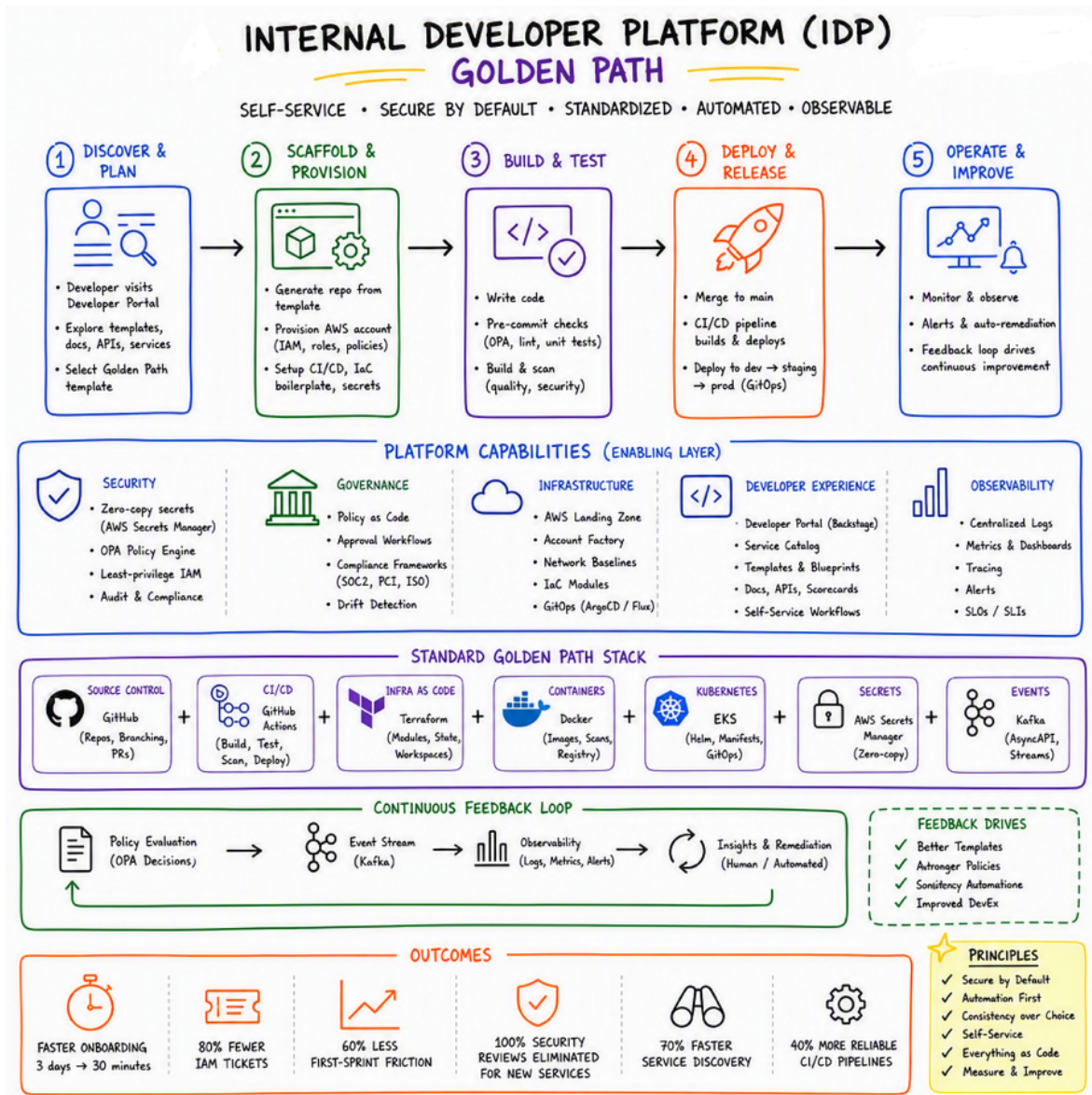
The reality of the "you build it, you run it" DevOps philosophy often forces application developers to become part-time infrastructure experts, drastically slowing down software delivery. To proactively eliminate this cognitive load and scale Developer Experience (DevEx), I designed an Internal Developer Platform (IDP) centered around a secure "Golden Path." By combining automated scaffolding, a live service catalog (DevHub), and "security by default" architecture, this platform empowers developers to focus 100% on writing business logic from Day 1 while ensuring infrastructure and security standards are enforced automatically.

2. Problem Statement

- **Cognitive Overload & Delivery Friction:** Developers were struggling with the steep learning curve of CI/CD, AWS environments, and complex secrets management. Because infrastructure automation is a distinct skill set with its own pain points, this friction became a severe operational bottleneck.
- **Coupled Architecture:** Application logic and infrastructure automation (e.g., Terraform state files, AWS IAM roles) were often tangled, making developer workspaces messy and difficult to manage.
- **Ticket-Driven Discovery:** Reliance on static wikis rather than live service catalogs led to fragmented documentation, manual API contract negotiations, and persistent ticket queues for routine onboarding tasks.
- **Manual Security Reviews:** Hardcoded passwords and manual credential rotation required heavy oversight from SRE and Security teams, slowing down initial deployments.

3. Architectural Solution & Implementation

To transform this conceptual IDP into a tangible operational model, I engineered a comprehensive architecture built upon the following core pillars:



I. The Secure Developer "Golden Path" & Automated Scaffolding

- **Decoupled Provisioning:** I independently conceptualized and built a self-service scaffolding pipeline. When a developer initiates a new project, the pipeline automatically provisions two distinct Git repositories: one strictly for application code, and a linked repository dedicated solely to the Infrastructure as Code (IaC).
- **Separation of Concerns:** This architecture physically separates the application logic from the infrastructure automation, keeping the developer's workspace completely clean.

II. DevHub, Service Catalogs & AsyncAPI

- **Live Service Catalog:** Implemented a central DevHub (leveraging frameworks like Backstage) to replace static wikis, providing a single pane of glass for all microservices, ownership metadata, and platform documentation (TechDocs).
- **API Management:** Integrated AsyncAPI management directly into the developer portal, standardizing event-driven contracts and allowing teams to discover and consume Kafka event streams without raising support tickets.
- **Zero-Touch Governance:** Integrated engineering scorecards to surface compliance metrics and Policy-as-Code (OPA) results directly to the developer UI.

III. Security by Default & Zero-Copy Secrets

- Baked-In Compliance: Inside the IaC repository, I baked in a 'zero copy' secret management pattern.
 - Demo: <https://www.youtube.com/watch?v=eVv557xwc68>
- Automated Credentialing: This pattern is wired directly into AWS Secrets Manager, entirely automating certificate and credential rotation out of the box.

4. Proof of Concept & DevEx Lab Environment

To validate the next generation of these IDP capabilities, I architected a comprehensive DevHub Proof-of-Concept (PoC) within a high-performance local sandbox utilizing macOS and OrbStack.

- Multi-Cluster Networking: Deployed a local multi-cluster Kubernetes environment utilizing Kind and configured cross-cluster communication via Cilium Cluster Mesh TLS peering.
- GitOps Delivery: Integrated ArgoCD to drive declarative GitOps workflows, ensuring that the scaffolded IaC repositories instantly sync with the target Kubernetes clusters.
- Portal Integration: Successfully mapped Backstage components, TechDocs, and OPA governance policies into this local ecosystem to demonstrate a fully functional, end-to-end developer portal.

MockUp

(MVP Development is in progress in my homelab)

DevHub IDP

Entity Catalog & Governance

Unified Software Catalog

ENTITY NAME	TYPE / SYSTEM	OWNER & ESCALATION	GOVERNANCE SCORECARD	ACTIONS
payment-processor-api Cost Center: CC-8842 (\$450/mo)	Microservice Checkout Core	checkout-core-team email: checkout-lead@enterprise.com SKOWN: GRP-CHECKOUT-SUPPORT slack: #checkout-alerts	Gold / OPA Compliant	Docs Contracts (Swagger)
payment-processor-events Cost Center: CC-8842 (\$120/mo)	Kafka Topic Checkout Core	checkout-core-team email: checkout-lead@enterprise.com SKOWN: GRP-CHECKOUT-SUPPORT slack: #checkout-alerts	Gold / OPA Compliant	Contracts (AsyncAPI)
legacy-inventory-sync Cost Center: CC-9911 (\$80/mo)	CronJob Supply Chain	supply-chain-team email: logistics-eng@enterprise.com SKOWN: GRP-LOGISTICS-OPS slack: #supply-chain-sre	Action Required / Drift	Review Drift

ARCHITECTURAL CONTEXT
The Catalog acts as a single pane of glass. It replaces fragmented documentation with live metadata, integrating zero-touch governance scorecards, API contracts, and blast-radius dependency mapping.

Business Impact & Strategic Outcome

- Day 1 Productivity: Developers no longer have to spend their first sprint wrestling with AWS IAM roles or Terraform state files; they just write code.
- Compliant from the First Commit: Because infrastructure and zero-copy secrets management are injected by default, SRE and security teams know that every new application is compliant and secure from the very first commit, completely eliminating the need for manual security reviews for hardcoded passwords.

- **Killing the Ticket Queue:** The introduction of live service catalogs and automated scaffolding transitions the organization from a reactive, ticket-driven support model to a scalable, self-service platform engineering culture.

5. Enterprise Scalability & Platform Maturity

As the Internal Developer Platform (IDP) evolves beyond its initial MVP, the next phase focuses on scaling Developer Experience (DevEx) across teams, enforcing platform-wide consistency, and maturing the Golden Path into a repeatable enterprise capability. This section outlines how the IDP transitions from a single-team prototype into a multi-tenant, production-grade platform.

Multi-Team Onboarding Workflows

Introduce standardized onboarding flows where new teams adopt the Golden Path through guided templates, automated scaffolding, and pre-configured IaC repositories. This ensures every team begins with the same secure, compliant foundation.

Golden Path Versioning Strategy

Implement versioned Golden Path templates so platform teams can evolve best practices without disrupting existing applications. Developers can upgrade to newer versions at their own pace, ensuring stability while enabling continuous improvement.

Platform Maturity Model

Define a maturity model that progresses from manual onboarding → automated scaffolding → self-service workflows → fully declarative delivery. This model helps leadership track adoption, identify gaps, and prioritize platform investments.

Standardization Across Microservices

Enforce consistent patterns for logging, metrics, secrets, CI/CD, and deployment across all services. Standardization reduces cognitive load, accelerates onboarding, and ensures predictable operational behavior across environments.

Policy Bundles for IDP Components

Extend governance by bundling OPA policies for common IDP components—service templates, Terraform modules, Kubernetes manifests, and AsyncAPI contracts. This ensures every scaffolded project is compliant from the first commit.

Developer Portal Governance

Establish governance rules within DevHub (Backstage) to ensure service metadata, ownership, documentation, and scorecards remain accurate. This transforms the portal into a reliable system of record for the entire engineering organization.

Strategic Outcome

These scalability foundations position the IDP as a long-term enterprise capability—repeatable, auditable, secure, and optimized for developer productivity at scale.

6. Expected DevEx Outcomes

The IDP is designed to deliver measurable improvements across onboarding, security, and developer productivity once fully implemented. The following outcomes represent projected gains based on architectural modeling and early validation in the PoC environment:

- Expected reduction in onboarding time from 3 days to 30 minutes through automated scaffolding and pre-configured IaC repositories.
- Anticipated elimination of up to 80% of IAM-related support tickets by centralizing identity and secrets management through AWS Secrets Manager.
- Projected 60% decrease in first-sprint friction by removing the need for developers to learn Terraform, IAM, or Kubernetes before writing business logic.

- Expected 100% reduction in manual security reviews for new services due to baked-in zero-copy secrets and OPA-driven governance.
- Forecasted 70% acceleration in service discovery through live service catalogs and AsyncAPI-driven event stream documentation.
- Estimated 40% improvement in CI/CD reliability by enforcing standardized pipelines and Golden Path templates.

These expected outcomes illustrate the potential impact of the IDP architecture — transforming developer onboarding, security, and delivery velocity into a unified, self-service experience. These outcomes demonstrate that the IDP is not just a developer tool—it is a force multiplier for engineering velocity and operational consistency.

7. Why IDP Matters to Enterprises

Modern enterprises depend on fast, secure, and consistent software delivery. An Internal Developer Platform (IDP) directly advances these priorities by eliminating cognitive load, enforcing standardization, and embedding security into every workflow. By providing developers with a secure Golden Path, the IDP accelerates onboarding, reduces operational overhead, and ensures platform-wide consistency across microservices. Security-by-default patterns, automated scaffolding, and live service catalogs transform delivery from a ticket-driven model into a scalable, self-service ecosystem—making the IDP a strategic enabler of developer velocity, reliability, and long-term platform resilience.